ELEVANCESKILLS

CRYPTOCURRENCY DATA ANALYSIS

Excel Dashboard Development Project

PROJECT REPORT
A detailed record of the development, analysis, dashboard design and outcomes of six cryptocurrency analytics tasks.

Organization

ElevanceSkills

Project focus: cryptocurrency market data, comparison, liquidity, price movement and investment insights.

Prepared as a consolidated project report

1. Executive Summary

This project involved building a collection of six Excel-based dashboards and analytical tools using cryptocurrency market data. The work focused on converting a dataset of approximately 200 crypto coins into clear, interactive views that help users compare assets, understand trading activity, examine price movements and explore price-based groups.

The dashboards use spreadsheet features such as formulas, data validation, sorting and filtering, PivotTables, slicers, charts and KPI-style summaries. A VBA macro was also used for the time-based visibility requirement in the first dashboard. The completed set is intended to demonstrate practical data preparation, spreadsheet analysis, dashboard construction and presentation skills.

2. Project Background and Objectives

Cryptocurrency market information is available in many fields, including current price, trading volume, market capitalization, circulating supply and percentage changes over different time periods. Looking at raw rows alone can make it difficult to identify useful patterns. This project addresses that problem by organizing the data into focused dashboards, each designed around a particular analytical question.

The main objectives were:

- Organize cryptocurrency market data so it can be filtered, compared and analyzed in Excel.
- Create clear charts and KPI summaries that communicate important values quickly.
- Build interactive filters and slicers so dashboard results respond to selected price categories.
- Compare trading volume, liquidity-related activity and price changes across selected coins.
- Apply spreadsheet formulas and VBA to introduce validation, calculations and time-based dashboard behavior.
- Present the results in a structured, understandable format suitable for review.

3. Dataset Description

The project used a cryptocurrency dataset containing approximately 200 coin records. The workbook included fields used throughout the tasks, such as coin name, symbol, price in USD, 24-hour trading volume, market capitalization, circulating supply and percentage changes over 1 hour, 24 hours and 7 days. The data was arranged in a tabular worksheet so that formulas, filters, PivotTables and charts could refer to consistent columns.

Data handling: Numeric fields such as price and trading volume were checked and formatted as numbers where necessary. This was important because values stored as text can cause incorrect sorting, calculations, chart series or PivotTable summaries. Helper columns and formulas were used where a task required derived values or categories.

4. Software and Excel Features Used

Tool / Feature	Purpose in the project
Microsoft Excel	Data organization, calculations, tables, dashboards and charts.
Formulas and functions	Validation, conditional logic, lookups, ranking, averages and derived price values.
PivotTables	Summarizing coin records and supporting interactive analysis.
Slicers and filters	Selecting price ranges and updating related dashboard views.

Tool / Feature	Purpose in the project
Charts	Presenting volume distribution, comparisons and price-change insights.
VBA macros	Controlling dashboard visibility according to the specified working-hour window and refreshing the dashboard.

5. Dashboard 1 — Time-Based Visibility and Sentiment Analysis

Purpose: To display the top 10 cryptocurrency coins by 24-hour trading volume while applying a name-based selection rule and a time-based display condition.

Implementation: The source data was filtered to include coin names beginning with a vowel or with B, C or D. The selected coins were ranked using the 24-hour volume field, and the top results were presented in a chart. The chart was designed to be visible only during the working-hour window from 9:00 AM to 5:00 PM. Outside this window, the workbook displays the message: “Please open in working hours (9 am to 5 pm).”

Excel and VBA work: A macro was added to control the chart's visibility and refresh the dashboard. The workbook was saved in macro-enabled format so that VBA functionality could be retained. The refresh procedure displays a confirmation message when the dashboard is refreshed.

Output: A volume-focused chart with a time-based visibility rule and a refresh action. The task demonstrates filtering, ranking, chart presentation and basic VBA automation.

6. Dashboard 2 — Crypto Coin Comparison Dashboard

Purpose: To compare two cryptocurrencies side by side using key market indicators.

Implementation: The dashboard accepts two coin names and retrieves their symbol, current price, 24-hour volume, market capitalization and circulating supply. Input validation was applied to reduce incorrect entries. The validation rule checks that a coin-name entry is between 3 and 10 characters and does not contain numeric digits.

Three KPI comparisons were included to show the difference between the selected coins in 24-hour volume, market capitalization and circulating supply. These comparisons make it easier to identify which of the two coins has the higher value for each measure.

Output: A two-coin comparison view with five market details for each coin and three difference KPIs. This task demonstrates input validation, lookup-based retrieval and comparative calculations.

7. Dashboard 3 — Top 5 Crypto Liquidity Analysis

Purpose: To show how 24-hour trading volume is distributed among the most active coins within a selected price category.

Implementation: A price-category selector divides coins into two groups: $0–$50 and above $50. Based on the selected category, the dashboard identifies the top five coins by 24-hour trading volume. All other coins in that category are combined into a single “Others” group. A pie chart presents the resulting distribution.

The chart and its supporting values are linked to the selected price category so that the analysis can change when the user selects a different category. The price and volume fields were treated as numeric values to support correct filtering, ranking and charting.

Output: An interactive top-five liquidity-style volume distribution chart with an “Others” category. The dashboard uses trading volume as an indicator of market activity; it should not be interpreted as a complete measure of market liquidity, which can also depend on factors such as bid–ask spread and order-book depth.

8. Dashboard 4 — Price Change Comparison Analysis

Purpose: To compare a cryptocurrency's current price with an estimated previous-hour price and identify the largest 1-hour percentage movements.

Implementation: The 1-hour percentage change was used to estimate the previous-hour price from the current price. Under the project's stated assumption that the reported 1-hour percentage represents a positive change, the estimated earlier price is calculated by reversing that percentage from the current price. The dashboard ranks coins by 1-hour price change and displays the top 10 results.

A price-range slicer separates coins into “Up to $10” and “Greater than or equal to $10” groups. The displayed symbol and price-change values are linked to the selected group so that the comparison responds to the slicer selection.

Output: A top-10 price-change comparison with a price-range filter. The previous-hour price is an estimate derived from the percentage-change field, not a separately observed historical market quote.

9. Dashboard 5 — Low Budget Investment Insights

Purpose: To help users explore lower-priced and higher-priced coin groups and identify the coin with the lowest average percentage change under the project's assumed downfall model.

Implementation: The dashboard treats the 1-hour, 24-hour and 7-day percentage-change fields as price-drop measures, as specified in the task instructions. It calculates an average downfall percentage for each coin using those three values. The dashboard then identifies the coin with the lowest average downfall in the selected price range.

Five price categories are provided: $0–$0.05, $0.05–$0.50, $0.50–$5, $5–$50 and above $50. The KPI area displays the selected coin's name, symbol, price and average downfall percentage, together with the total number of coins in the selected category. The total count is intended to update with the slicer selection.

Output: A KPI-based dashboard for comparing coins across five price bands. The result is an analytical screening view based on the assumptions in the task; it is not personalized investment advice or a prediction of future performance.

10. Dashboard 6 — Historical Price Change Insights

Purpose: To compare estimated prices across different time periods for coins priced between $0 and $5, with a focus on the top 10 coins ranked by 1-hour percentage change.

Implementation: The dataset was filtered to the $0–$5 price range. The 1-hour percentage change was used to rank the coins, and the top 10 were selected for the chart. The task applies the stated assumptions that the 1-hour and 7-day changes represent increases while the 24-hour change represents a decrease. Using these assumptions, estimated comparison prices can be derived from the current price and the corresponding percentage-change fields.

The chart compares the estimated 7-day and 24-hour price values for the selected top 10 coins. The calculations depend on the sign convention and assumptions specified in the task, so the derived values should be understood as estimates rather than verified historical closing prices.

Output: A filtered top-10 comparison chart showing estimated price levels for the two time periods. This task demonstrates filtering, ranking, percentage-based calculations and chart-based comparison.

11. Overall Project Methodology

The six dashboards were developed using a consistent workflow. First, the source worksheet was reviewed to identify the relevant columns for each task. Numeric fields were checked and formatted to avoid text-versus-number issues. Next, helper calculations, category labels or lookup formulas were created where required. The dashboard view was then built using charts, KPI cells, PivotTables, slicers or VBA, depending on the task. Finally, selections and outputs were checked by changing input values, price categories or slicer settings.

The general workflow can be summarized as follows:

- Review the source dataset and identify the fields required for the task.
- Clean or format values so that prices, volumes and percentage changes can be calculated correctly.
- Create formulas and helper fields for validation, ranking, grouping or derived prices.
- Build the required chart, KPI section, comparison table or PivotTable.
- Connect slicers, filters, input cells or VBA actions to the dashboard output.
- Test multiple selections and verify that displayed values update as expected.

12. Skills and Learning Outcomes

The project provided practical experience in using Excel for data analysis and dashboard development. It also highlighted the importance of checking source data types, understanding percentage calculations and testing interactive components rather than relying only on the appearance of a chart.

- Data preparation: organizing records and ensuring numeric values are suitable for calculations.
- Excel formulas: using logical tests, lookups, conditional calculations, averages and ranking logic.
- Data visualization: selecting charts that communicate comparisons and distributions clearly.
- Interactive reporting: using filters and slicers to let users explore different data groups.
- Automation: using VBA to control chart visibility and refresh behavior.
- Analytical reasoning: translating written task requirements into data fields, calculations and dashboard outputs.
- Quality checking: testing edge cases, selected categories, totals and the relationship between source data and dashboard results.

13. Project Limitations and Interpretation

The dashboards are based on the dataset available for this project. Cryptocurrency prices and trading volumes change continuously, so the results represent the values contained in the workbook at the time the data was collected or prepared. The project does not establish that the dataset is a live market feed unless it is refreshed from a current data source.

Some tasks explicitly require assumptions about whether percentage changes represent increases or decreases. Calculated previous-period prices are therefore estimates based on those assumptions and the supplied percentage-change fields. Similarly, 24-hour volume is used as a practical activity measure in the liquidity dashboard, but volume alone does not capture every dimension of liquidity.

The dashboards are educational and analytical tools. They do not guarantee investment returns, forecast future prices or provide individualized financial recommendations.

14. Conclusion

The cryptocurrency data analysis project for ElevanceSkills brings together six Excel dashboards covering time-based volume visibility, coin comparison, top-five volume distribution, price-change comparison, low-budget coin screening and historical price-change insights. Across these tasks, the project applies data preparation, formulas, charts, KPI summaries, slicers, PivotTables and VBA to turn a raw cryptocurrency dataset into structured analytical views.

Completing the dashboards strengthened practical understanding of spreadsheet-based analysis and interactive reporting. The overall project demonstrates how Excel can be used to organize market data, calculate useful indicators and communicate findings through dashboards that are easier to explore than a raw data table.

END OF PROJECT REPORT